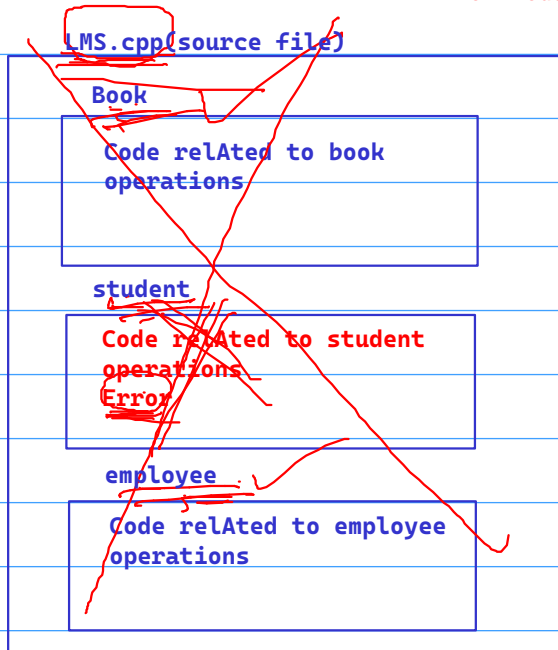


Library Mgmt System

Non Modular system



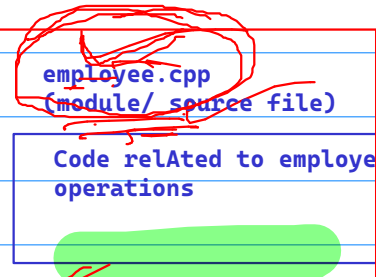
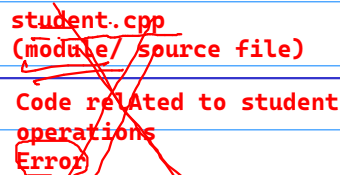
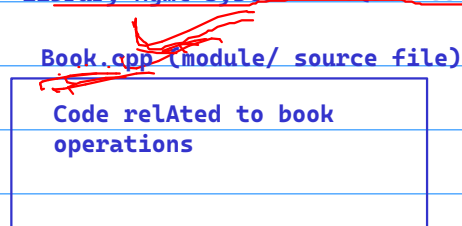
Here we have writren whole the code in single file

- * If 1 module fails entire system will fail
- * Dificutl to find an error and fix it

.cpp or .c $\Rightarrow$ source file or module

Modular approach

Library Mgmt System LMS (source file)



I have written the code in different files

Here if one mode fials then only that module can be kept down but whole program will not fail(kept down)

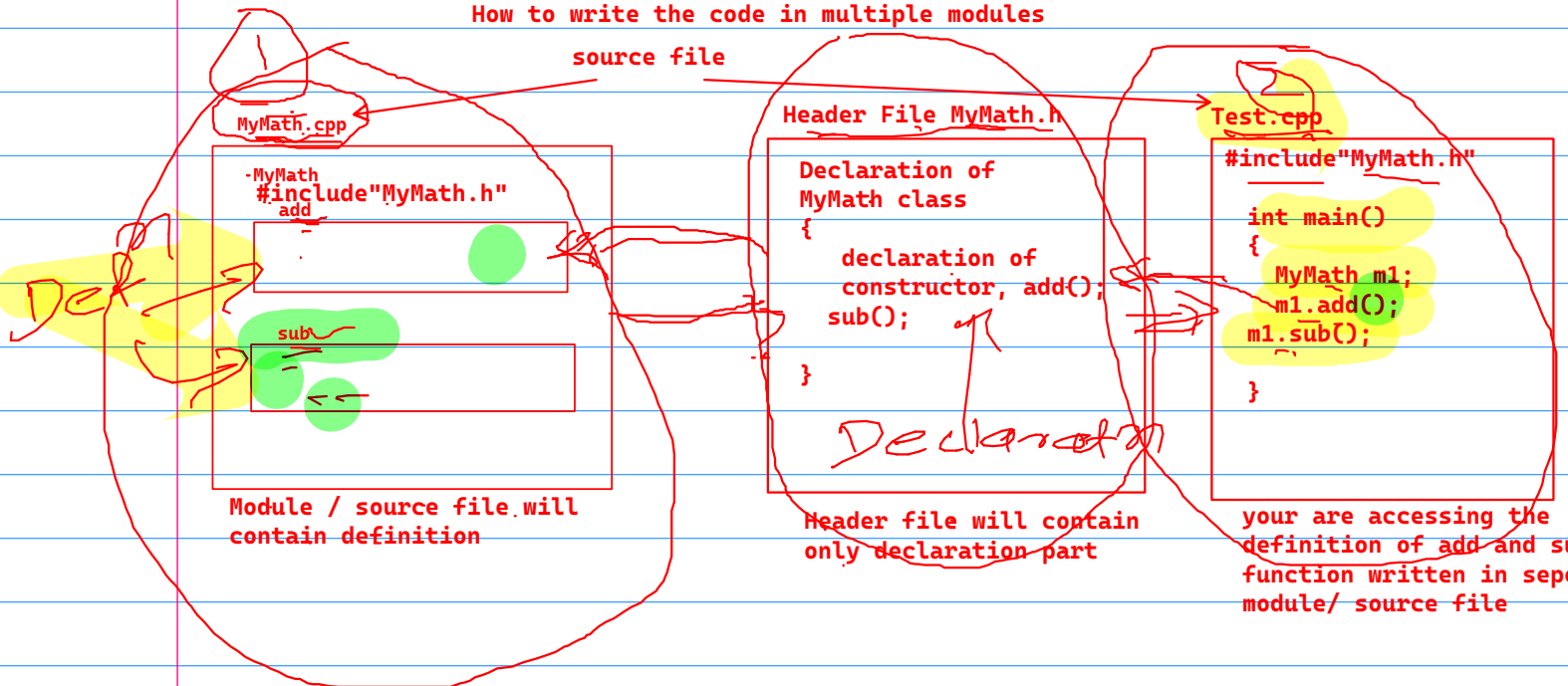
Employee MGMT System

employee module

Here earned the code reusability

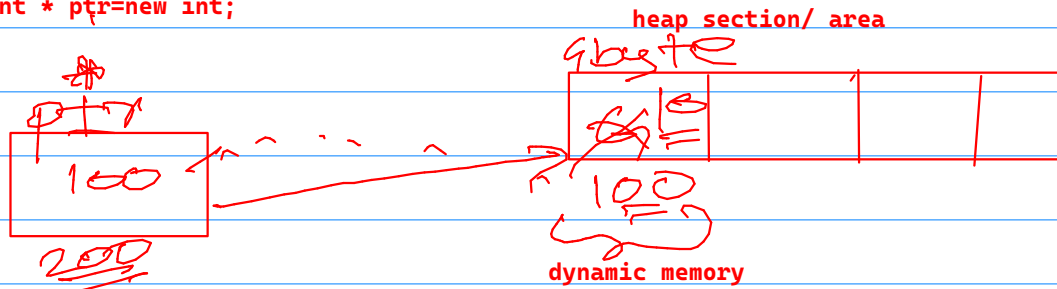
include

How to write the code in multiple modules



Dynamic memory allocation in C++

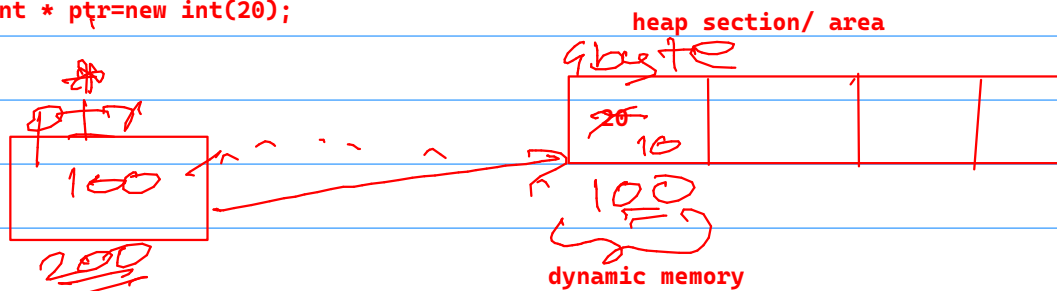
```
int * ptr=new int;
```



```
*ptr=10; //assignment
cout<<*ptr; //dereferencing operation to get the value pointer ptr
```

Dynamic memory allocation in C++

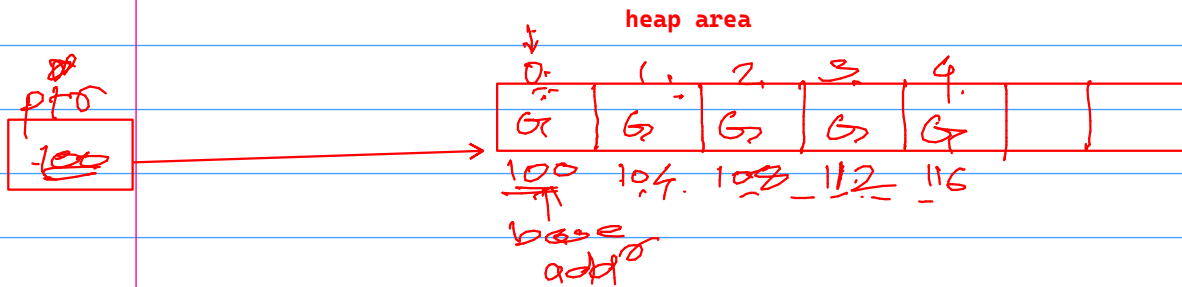
```
int * ptr=new int(20);
```



```
*ptr=10; //assignment
cout<<*ptr; //dereferencing operation to get the value pointer ptr
```

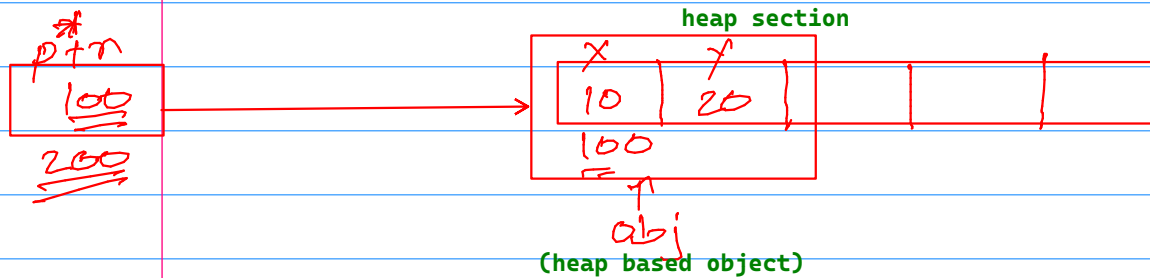
//In cpp allocating the dynamic memory for dynamic array(dynamic block)

```
int * ptr=new int[5]; // here 5*4 ⇒ 20 byte of dynamic memory will be allocated
//in heap section and initialized with garbage value
```



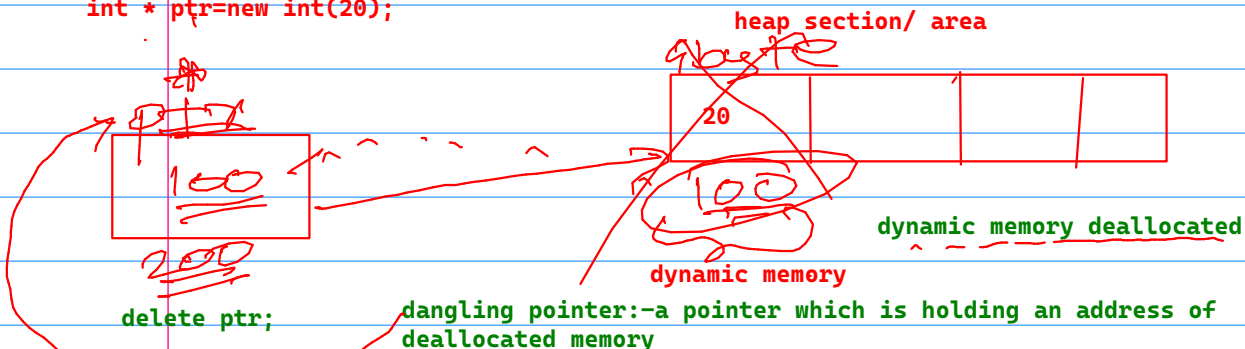
Heap based object

```
point * ptr=new point; //dynamic memory allocation will be done for point class object
//by programmer hence it is job programmer to deallocate the memory
```

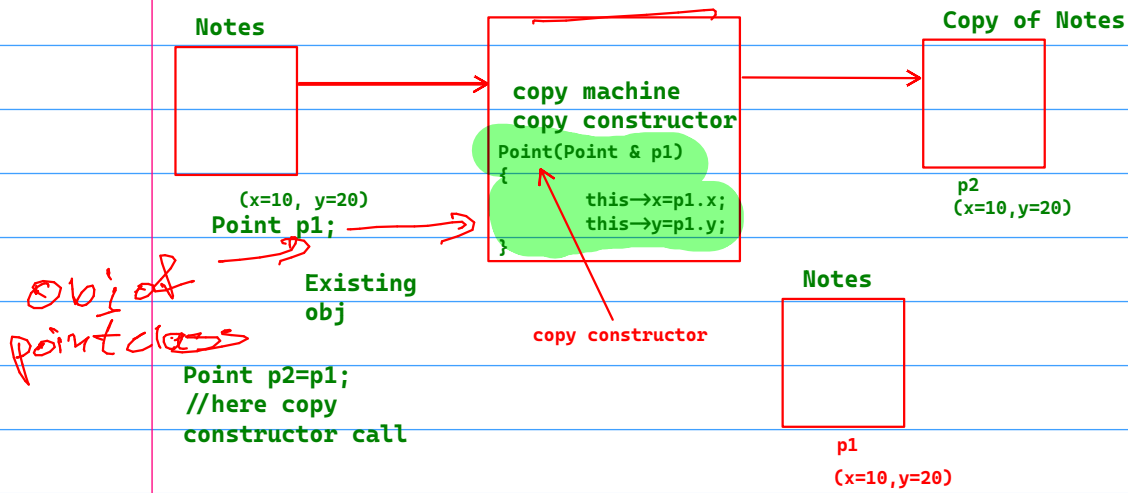


dangling pointer

```
int * ptr=new int(20);
```



Copy Constructor



class Person

```
name  
add  
date dob; //obj of date
```

here dob is an object of date class
which will be treated as data member
of Person class

"has a"

class date

```
int dd;  
int mm;  
int yy;
```

print()